ANALYSIS REPORT

RCIM Model-Bank Reproduction Full-Matrix Family Reproduction Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the full-matrix family-reproduction campaign prepared in:

- `doc/reports/campaign_plans/track_1/exact_paper/2026-04-14-13-42-10_track1_full_matrix_family_reproduction_campaign_plan_report.md`
  The campaign executed `20` exact-paper validation runs through the dedicated launcher:
- completed runs: `20`
- failed runs: `0`
- execution window: `2026-04-14 14:12:14+02:00` to `2026-04-14 14:15:18+02:00`
- campaign artifact root: `output/training_campaigns/track1/exact_paper/forward/full_matrix/shared/track1_full_matrix_family_reproduction_campaign_2026_04_14_13_50_51/` This campaign was designed for the clarified first objective of `RCIM Model-Bank Reproduction`:
- reproduce the paper matrices row by row with repository-owned model implementations;
- execute one amplitude run and one phase run for each paper family;
- materialize the real repository-side counterparts of paper Tables `3`, `4`, and `5`.

Objective And Outcome

The campaign had three concrete goals

1. prepare one exact-paper row-reproduction run for each paper family on the amplitude side;
2. prepare one exact-paper row-reproduction run for each paper family on the phase side;
3. replace the old best-envelope reading with a real full-matrix repository-owned replication surface. Outcome:

- all `20` runs completed successfully;
- all `20` runs exported ONNX artifacts with `0` failed targets;
- the campaign produced the first complete repository-owned row-by-row replication surface for Tables `3` , `4` , and `5` ;
- `RCIM Model-Bank Reproduction` remains open because no table is fully green and several rows still contain materially red cells;
- the strongest row-level reproductions are now explicit and inspectable rather than hidden inside one all-family winner run. This means the campaign succeeded operationally and scientifically as a matrix-reproduction pass, even though it did not close `RCIM Model-Bank Reproduction` .

Ranking Policy

This campaign cannot be ranked by one global `MAPE` winner, because each run only reproduces one paper family row on one target side.

The serialized campaign policy is therefore

- primary metric: `row_closure_score_desc`
- first tie breaker: `met_paper_cell_count_desc`
- second tie breaker: `open_paper_cell_count_asc`
- third tie breaker: `mean_normalized_gap_ratio_asc`
- fourth tie breaker: `max_normalized_gap_ratio_asc`
- fifth tie breaker: `matched_table6_family_target_count_desc`
- sixth tie breaker: `run_name`

  Where:

- `row_closure_score = (met + 0.5 * near) / target_count`
- `met` means the repository row cell reached or beat the paper cell
- `near` means the positive gap stayed within `25%` of the paper cell

  Interpretation:

- this policy exists only to serialize the required campaign-best artifacts;
- the real scientific interpretation remains full matrix replication, not one global best run.

Campaign Ranking

Ranked Completed Runs

Rank	Run	Family	Scope	Targets	Met	Near	Open	ClosureScore
1	track1_rf_phase_full_matrix	RF	phases_only	9	3	4	2	0.556
2	track1_ert_amplitude_full_matrix	ERT	amplitudes_only	10	3	4	3	0.500
3	track1_gbm_phase_full_matrix	GBM	phases_only	9	3	3	3	0.500
4	track1_hgbm_amplitude_full_matrix	HGBM	amplitudes_only	10	3	4	3	0.500
5	track1_hgbm_phase_full_matrix	HGBM	phases_only	9	2	5	2	0.500
6	track1_rf_amplitude_full_matrix	RF	amplitudes_only	10	2	5	3	0.450
7	track1_ert_phase_full_matrix	ERT	phases_only	9	3	2	4	0.444
8	track1_lgbm_amplitude_full_matrix	LGBM	amplitudes_only	10	2	4	4	0.400
9	track1_lgbm_phase_full_matrix	LGBM	phases_only	9	1	5	3	0.389
10	track1_gbm_amplitude_full_matrix	GBM	amplitudes_only	10	1	4	5	0.300
11	track1_dt_phase_full_matrix	DT	phases_only	9	0	4	5	0.222
12	track1_dt_amplitude_full_matrix	DT	amplitudes_only	10	0	2	8	0.100
13	track1_et_amplitude_full_matrix	ET	amplitudes_only	10	0	2	8	0.100
14	track1_et_phase_full_matrix	ET	phases_only	9	0	1	8	0.056
15	track1_xgbm_amplitude_full_matrix	XGBM	amplitudes_only	10	0	1	9	0.050
16	track1_svm_amplitude_full_matrix	SVR	amplitudes_only	10	0	1	9	0.050
17	track1_xgbm_phase_full_matrix	XGBM	phases_only	9	0	0	9	0.000
18	track1_svm_phase_full_matrix	SVR	phases_only	9	0	0	9	0.000
19	track1_mlp_phase_full_matrix	MLP	phases_only	9	0	0	9	0.000
20	track1_mlp_amplitude_full_matrix	MLP	amplitudes_only	10	0	0	10	0.000

Export Surface

Completed Runs	Exported ONNX Files	Failed Exports	Surrogate Exports
20	190	0	4

The `4` surrogate exports all came from the `SVR` amplitude row, where a few targets still needed the existing constant-linear-regression safeguard.

Campaign Best Run

The explicit bookkeeping winner is

- `track1_rf_phase_full_matrix`

  It was selected because it achieved:

- the highest campaign row-closure score: `0.556`
- `3` met cells and `4` near cells out of `9`
- only `2` materially open cells in the `RF` phase row

  Important interpretation:

- this run is the campaign bookkeeping winner;
- it is **not** the global solution for `RCIM Model-Bank Reproduction`;
- it only means the `RF` phase row is currently the cleanest single row in the full-matrix campaign.

Matrix-Reproduction Impact

Campaign-Wide Cell Totals

Surface	Green	Yellow	Red
Table `3` amplitude `RMSE`	11	27	62
Table `4` phase `MAE`	16	28	46
Table `5` phase `RMSE`	16	22	52

Strongest Current Rows

- strongest amplitude-side row: `ERT` with `3` green, `4` yellow, `3` red
- strongest phase-side combined row: `RF` with `3` met, `4` near, `2` open
- strongest phase `MAE` row by direct visual usefulness: `RF`
- strongest phase `RMSE` rows: `RF` and `HGBM`

Weakest Current Rows

- weakest amplitude row: `MLP` with `0` green, `0` yellow, `10` red
- weakest phase rows: `SVM`, `XGBM`, and `MLP`, all still fully open under the paired row-closure reading

Interpretation By Table

Table 3 - Amplitude RMSE

The amplitude side is not uniformly hard.

The campaign shows three distinct bands

- strongest rows: `ERT` , `HGBM` , `RF`
- usable but incomplete rows: `LGBM` , `GBM`
- clearly off-paper rows: `SVM` , `DT` , `ET` , `XGBM` , `MLP`

Important reading:

- the amplitude side is now clearly tree and boosting dominated;
- `ERT` and `HGBM` are the most paper-faithful amplitude families overall;
- `MLP` is not remotely competitive on this exact-paper branch.

Table 4 - Phase MAE

Phase `MAE` is the healthiest matrix of the three.

The best rows are

- `RF`
- `ERT`
- `GBM`
- `HGBM`

Important reading:

- several families are already green on the mid-order harmonics;
- the persistent blockers are still concentrated at `162` and `240` ;
- `RF` is the cleanest phase-side row under the current exact-paper split.

Table 5 - Phase RMSE

Phase `RMSE` is materially harder than phase `MAE` .

The best rows are

- `RF`
- `HGBM`
- `GBM`
- `ERT`

Important reading:

- `78` , `81` , and `156` are already strong across several rows;
- the dominant unresolved columns are `162` and `240` ;
- no family currently gives a paper-faithful fully green phase- `RMSE` row.

Main Conclusions

The campaign supports six conclusions.

1. The Full-Matrix Objective Is Now Operationally Real

This is the first repository-owned campaign that truly reproduces the paper matrix structure family by family.

That removes the ambiguity of the previous best-envelope reading.

2. RCIM Model-Bank Reproduction Is Still Open

No table is fully green and no family row is fully replicated.

So the first true `RCIM Model-Bank Reproduction` objective is not yet closed.

3. Tree And Boosting Families Dominate The Exact-Paper Surface

The strongest rows come from

- `RF`
- `ERT`
- `GBM`
- `HGBM`
- `LGBM` on selected rows

This is directionally consistent with the paper.

4. The Phase Side Is Closer Than The Amplitude Side

The campaign-wide totals show

- amplitude still has `62` red cells
- phase `MAE` has `46` red cells
- phase `RMSE` has `52` red cells

So the amplitude matrix remains the larger closure burden.

5. MLP Is The Clearest Non-Starter On This Branch

Both `MLP` rows remain scientifically weak.

That does not just mean â€œnot bestâ€�; it means materially off the paper matrix.

6. The Next Iteration Should Be Row-Aware And Harmonic-Aware

This campaign was the correct first packaging step.

But the next one should be more surgical:

- keep the stronger families for the rows they already serve well;
- focus targeted repair work on the remaining red harmonics within those rows;
- avoid spending equal effort on rows that are structurally far from the paper.

Recommended Next Actions

The next technically justified steps are

1. keep `track1_rf_phase_full_matrix` as the bookkeeping winner for this campaign only;
2. keep the real `RCIM Model-Bank Reproduction` reading anchored to the full family-row matrices in the benchmark report;
3. prioritize row-aware repair work around: `ERT` amplitude, `HGBM` amplitude, `RF` phase, `GBM` phase, and `HGBM` phase;
4. deprioritize `MLP` and likely also `SVM` / `XGBM` for immediate closure work unless there is a strong paper-specific reason to keep them active;
5. use the new matrix dashboard to choose the next repair campaign by red-cell density rather than by generic family subset intuition.

Artifact References

- Campaign root: `output/training_campaigns/track1/exact_paper/forward/full_matrix/shared/track1_full_matrix_family_reproduction_campaign_2026_04_14_13_50_51/`
- Campaign leaderboard: `output/training_campaigns/track1/exact_paper/forward/full_matrix/shared/track1_full_matrix_family_reproduction_campaign_2026_04_14_13_50_51/campaign_leaderboard.yaml`
- Campaign best run YAML: `output/training_campaigns/track1/exact_paper/forward/full_matrix/shared/track1_full_matrix_family_reproduction_campaign_2026_04_14_13_50_51/campaign_best_run.yaml`
- Campaign best run note: `output/training_campaigns/track1/exact_paper/forward/full_matrix/shared/track1_full_matrix_family_reproduction_campaign_2026_04_14_13_50_51/campaign_best_run.md`
- Campaign logs: `output/training_campaigns/track1/exact_paper/forward/full_matrix/shared/track1_full_matrix_family_reproduction_campaign_2026_04_14_13_50_51/logs/`
- Canonical benchmark dashboard: `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md`